

Temporal Topic Modeling and Citation Recommendation for Scientific Literature Using LDA and Hawkes Processes

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ABSTRACT

Scientific literature in artificial intelligence and machine learning is growing rapidly, making it difficult for researchers to track emerging topics and identify relevant prior work. Traditional topic models such as Latent Dirichlet Allocation (LDA) provide only static representations of topical structure and fail to capture temporal influence patterns in scientific publishing. In this work, we propose a two-stage framework that combines frozen LDA topic representations with a Hawkes process-based temporal model augmented by citation-supervised attention and a Weibull temporal kernel. The attention mechanism learns which past papers are most relevant to the current research context using citation relationships, while the Weibull kernel models delayed citation influence more realistically than standard exponential decay. We evaluate the framework on arXiv papers from the cs.CL and cs.LG categories (2014–2021) for topic prediction and citation recommendation. Experimental results show that our model achieves the best citation recommendation performance with an AUC-ROC of 0.845 while also producing more coherent topics than Dynamic Structured Neural Topic Models (DSNTM). The proposed framework demonstrates the effectiveness of combining topic modeling with continuous-time temporal influence learning for scientific trend analysis and citation recommendation.

I. INTRODUCTION

Scientific publications in artificial intelligence (AI) and machine learning (ML) are growing at an unprecedented rate. Platforms such as arXiv receive thousands of new submissions daily across subfields including natural language processing (NLP), computer vision, and reinforcement learning. This exponential growth creates two fundamental challenges for researchers: (1) tracking which research topics are emerging

or declining, and (2) identifying the most relevant prior work to cite when writing a new paper.

Traditional topic models such as Latent Dirichlet Allocation (LDA) [1] provide a static snapshot of the topical structure in a corpus. Given a collection of documents, LDA discovers K latent topics, each represented as a distribution over the vocabulary. While powerful for exploratory analysis, LDA treats all documents as exchangeable—a paper from 2014 and one from 2021 are treated identically. This static view fundamentally cannot capture how topics evolve over time or how one influential paper triggers a cascade of related work.

Scientific publishing exhibits a *self-exciting* behavior: a breakthrough paper (e.g., “Attention Is All You Need” [5]) triggers a wave of follow-up research, and each follow-up paper can itself trigger further work. This cascading temporal influence is precisely the phenomenon modeled by the *Hawkes process* [2]—a self-exciting point process where the occurrence of past events increases the probability of future events.

In this work, we propose a framework that bridges topic modeling and temporal influence learning for scientific trend prediction and citation recommendation. Our approach separates topic discovery (handled by frozen LDA) from temporal dynamics (handled by a Hawkes process with a Weibull kernel and citation-supervised attention). This separation improves interpretability, training stability, and enables the model to perform both topic prediction and paper-level citation recommendation—a task that existing dynamic topic models like DSNTM [4] cannot address.

Our key contributions are:

- 1) A two-stage LDA + Hawkes framework that separates static topic discovery from continuous-time temporal modeling.
- 2) A Weibull temporal kernel that captures delayed citation peaks, replacing the unrealistic exponential decay used in prior Hawkes-based topic models.
- 3) A citation-supervised attention mechanism that learns which past papers are relevant to the current research context, grounded in actual citation behavior via BCE loss.

- 4) A context vector formulation $\mathbf{c}(T^*)$ that ensures consistent behavior during both training and inference.
- 5) Evaluation on two tasks-topic prediction and citation recommendation-demonstrating that our model achieves the best AUC-ROC (0.845) for citation recommendation while producing the most coherent topics (NPMI 0.055).

II. RELATED WORK

A. Hawkes Topic Model (HTM)

The Hawkes Topic Model (HTM) [3] was among the first to combine topic modeling with multivariate Hawkes processes. HTM jointly learns latent topics and a hidden diffusion network, modeling how publications by one author trigger publications by another on similar topics. The model uses variational inference for joint optimization and employs an exponential temporal decay kernel.

While HTM elegantly captures influence diffusion dynamics and has a strong probabilistic foundation, it has several limitations for our task. First, it was designed for *network inference* (discovering "who influenced whom"), not for *topic prediction* (forecasting "what emerges next"). Second, the exponential decay kernel $g(\Delta t) = \alpha \cdot e^{-\beta \Delta t}$ assumes influence is strongest immediately after publication and monotonically decreases—a poor fit for citation dynamics, where influence typically peaks weeks to months after publication. Third, HTM lacks any attention mechanism for understanding semantic relevance between papers.

B. Dynamic Structured Neural Topic Model (DSNTM)

The Dynamic Structured Neural Topic Model (DSNTM) [4] is a neural topic model that captures topic evolution across time using self-attention over topic embeddings. As a VAE-based generative model, DSNTM learns time-varying topic-word distributions $\beta_k^{(t)}$, allowing the vocabulary associated with each topic to evolve across discrete time bins. Crucially, DSNTM incorporates citation information through a citation regularization loss that pushes attention weights to reflect actual citation relationships—directly parallel to our approach.

DSNTM achieves strong performance on perplexity and topic coherence benchmarks and represents the state of the art for dynamic topic modeling on scientific corpora. However, it has three key limitations. First, it discretizes time into bins (e.g., years), losing fine-grained temporal information: a paper from January and one from December of the same year are treated identically. Second, it does not explicitly model the self-exciting cascade effect central to Hawkes processes—where paper A triggers paper B, which triggers paper C. Third, its joint optimization of topics and temporal dynamics increases training complexity.

DSNTM is our primary baseline due to its direct task alignment: it performs topic prediction, uses citation supervision, employs attention, and is evaluated on scientific corpora.

C. Other Related Work

Classical dynamic topic models such as the Dynamic Topic Model (DTM) [6] and the Continuous-Time Dynamic Topic

Model (cDTM) allow topics to evolve over time but do not incorporate citation information or self-exciting temporal dynamics. Neural topic models such as the Embedded Topic Model (ETM) and D-ETM extend LDA with neural components but similarly lack temporal influence modeling.

III. BACKGROUND: HAWKES PROCESSES

A Hawkes process [2] is a self-exciting point process defined by its conditional intensity function. For a univariate process, the intensity at time t is:

$$\lambda(t) = \mu + \sum_{e: t_e < t} g(t - t_e) \quad (1)$$

where $\mu > 0$ is the base intensity (background rate of events occurring without past excitation) and $g(\cdot)$ is the triggering kernel that determines how much each past event e at time t_e increases the current intensity.

Intuition. Consider earthquake modeling, the original motivation for Hawkes processes. A main shock (large earthquake) triggers aftershocks, which can trigger further aftershocks. In our domain, a landmark paper triggers follow-up papers, which themselves trigger more work—the same self-exciting cascade.

In the topic-specific formulation used by HTM [3], the intensity for topic k at time t is:

$$\lambda_k(t) = \mu_k + \sum_{e: t_e < t} \alpha_{e,k} \cdot f(t - t_e) \quad (2)$$

where μ_k is the base rate for topic k , $\alpha_{e,k}$ is the excitation weight (how strongly past paper e activates topic k), and $f(\cdot)$ is the temporal decay kernel.

Standard exponential kernel. Most existing Hawkes models use exponential decay:

$$g(\Delta t) = \alpha \cdot e^{-\beta \Delta t} \quad (3)$$

This is highest at $\Delta t = 0$ and strictly decreasing. However, in scientific publishing, citation influence typically rises gradually after publication, peaks after weeks or months, and then slowly declines. Exponential decay fundamentally cannot capture this delayed-peak behavior, motivating our use of the Weibull kernel described in Section IV-B3.

IV. PROPOSED FRAMEWORK

Our framework is a two-stage pipeline: Stage 1 trains LDA to produce frozen topic representations, and Stage 2 trains a Hawkes process with attention and Weibull decay over these frozen representations. At inference time (Stage 3), the trained model predicts future topic distributions and recommends citations.

A. Stage 1: LDA Pre-Training

We train a standard LDA model [1] using Gensim on the preprocessed arXiv abstracts with $K = 30$ topics. This produces two representations that are **frozen** after training:

- $\beta_k \in \mathbb{R}^V$ - the word-topic distribution for topic k ($K \times V$ matrix), where $\beta_{k,w} = P(w | k)$.

- $\theta_d \in \mathbb{R}^K$ - the document-topic proportions for paper d ($D \times K$ matrix), where $\theta_{d,k}$ is the proportion of topic k in paper d .

Design rationale. By freezing β and θ , we separate *what topics mean* (LDA’s responsibility) from *when and how topics activate* (the Hawkes model’s responsibility). This separation has three advantages: (1) it preserves clean, interpretable topics that joint training might distort, (2) it simplifies training by reducing the parameter space of the Hawkes model, and (3) it allows us to use well-established LDA implementations for topic discovery.

B. Stage 2: Hawkes Training

1) *Our Intensity Function:* The intensity of topic k at time T^* in our model is:

$$\lambda_k(T^*) = \mu_k + \sum_{e: t_e < T^*} \theta_e[k] \cdot A_{e,\text{ctx}} \cdot f_\Delta(T^* - t_e) \quad (4)$$

where μ_k is the learned base rate for topic k , and each past paper e contributes through three multiplicative factors:

- 1) **Topic weight** $\theta_e[k]$: The proportion of topic k in paper e (from frozen LDA). An NLP paper contributes strongly to NLP topic intensity but weakly to computer vision.
- 2) **Attention weight** $A_{e,\text{ctx}}$: A learned weight determining how relevant paper e is to the current research context (Section IV-B2).
- 3) **Weibull decay** $f_\Delta(T^* - t_e)$: The temporal influence of paper e given the time elapsed since its publication (Section IV-B3).

A past paper contributes meaningfully to topic k ’s intensity only if all three factors are non-negligible: the paper must be *about* topic k ($\theta_e[k]$ is high), *relevant* to the current context ($A_{e,\text{ctx}}$ is high), and published at an *appropriate time* (f_Δ is high).

2) *Attention Mechanism:* The attention module determines which past papers are most influential for future topic prediction:

$$A_{e,\text{ctx}} = \text{softmax} \left(\frac{\mathbf{c}(T^*)^\top \mathbf{W}_Q \cdot \mathbf{W}_K \theta_e}{\sqrt{K}} \right) \quad (5)$$

where $\mathbf{W}_Q, \mathbf{W}_K \in \mathbb{R}^{K \times K}$ are learned query and key projection matrices, and $\sqrt{K}$ provides scaling to prevent large dot products (following standard scaled dot-product attention [5]).

The **context vector** $\mathbf{c}(T^*)$ represents the current state of the research field:

$$\mathbf{c}(T^*) = \frac{1}{|\mathcal{W}|} \sum_{d \in \mathcal{W}} \theta_d, \quad \mathcal{W} = \{d : t_d \in [T^* - w, T^*]\} \quad (6)$$

where $\mathcal{W}$ is the set of papers in a recent temporal window of width w . If $\mathcal{W} = \emptyset$, we fall back to a uniform vector $(1/K, \dots, 1/K)$.

Why $\mathbf{c}(T^*)$ instead of θ_d as query? At inference time, the future paper does not exist—there is no θ_d to use. Using the context vector (mean of recent papers) ensures that the query distribution seen during training matches the one used at inference, preventing train-test mismatch.

Asymmetric relationships. The learned projection matrices $\mathbf{W}_Q$ and $\mathbf{W}_K$ enable the attention to capture asymmetric cross-topic influences. For example, the model can learn that “reinforcement learning papers tend to cite optimization papers” even without direct topic overlap between query and key vectors.

3) *Weibull Temporal Kernel:* We replace the standard exponential decay with a Weibull kernel:

$$f_\Delta(\Delta t; k_w, \lambda_w) = \frac{k_w}{\lambda_w} \left(\frac{\Delta t}{\lambda_w} \right)^{k_w-1} \exp \left(- \left(\frac{\Delta t}{\lambda_w} \right)^{k_w} \right) \quad (7)$$

where $\Delta t = T^* - t_e$ is the time elapsed since paper e was published, and:

- k_w (**shape parameter**): Controls how influence builds over time.
 - $k_w < 1$: Influence peaks immediately at $\Delta t = 0$ and monotonically decreases (similar to exponential decay).
 - $k_w = 1$: Reduces to exponential decay $f(\Delta t) = \frac{1}{\lambda_w} e^{-\Delta t/\lambda_w}$.
 - $k_w > 1$: Influence starts at zero, *rises* to a peak after a delay, then decays. This is the most realistic regime for citation dynamics.
- λ_w (**scale parameter**): Controls the duration of influence. Small λ_w produces short-lived bursts; large λ_w produces long-lasting influence.

Both parameters are learned jointly during training. The Weibull kernel with $k_w > 1$ naturally captures the “rise, peak, decay” pattern of citation influence—a paper receives few citations immediately after publication, reaches peak citations weeks to months later, and then gradually declines.

4) *Training Objective:* The model is trained with a multi-task loss:

$$\mathcal{L} = \mathcal{L}_{\text{Hawkes}} + \gamma \cdot \mathcal{L}_{\text{cite}} \quad (8)$$

Hawkes log-likelihood:

$$\mathcal{L}_{\text{Hawkes}} = \sum_d \left[\sum_k \log \lambda_k(t_d) - \int_0^T \lambda_k(t) dt \right] \quad (9)$$

The first term rewards high intensity at times when papers were actually published. The second term (the compensator) penalizes high intensity at times when no papers appeared. Together, they train the model to have high intensity precisely at observed events and low intensity elsewhere. The compensator integral is approximated via Monte Carlo sampling.

Citation regularization:

$$\mathcal{L}_{\text{cite}} = \sum_{d,e} \text{BCE}(A_{e,d}, C_{e,d}) \quad (10)$$

where $C_{e,d} = 1$ if paper e cites paper d (and 0 otherwise), and BCE is binary cross-entropy. This loss pushes the attention weights to align with the actual citation graph, grounding the learned attention in real academic influence patterns.

The hyperparameter γ balances the two losses. If γ is too small, citation information is underutilized; if too large, the temporal model is overwhelmed by citation supervision.

C. Stage 3: Prediction at T^*

At inference time, given a target time T^* (e.g., the year 2021):

Step 1: Context vector. Compute $\mathbf{c}(T^*)$ as the mean θ of papers in $[T^* - w, T^*]$.

Step 2: Topic intensities. Compute $\lambda_k(T^*)$ for all K topics using Eq. (4).

Step 3: Topic proportions. Convert intensities to a topic distribution via softmax:

$$\pi_k = \text{softmax}(\lambda(T^*))_k = \frac{e^{\lambda_k(T^*)}}{\sum_{j=1}^K e^{\lambda_j(T^*)}} \quad (11)$$

Step 4: Word distribution. Compute the predicted word distribution as a weighted mixture of frozen LDA topic-word distributions:

$$p(w \mid T^*) = \sum_{k=1}^K \pi_k \cdot \beta_{k,w} \quad (12)$$

Step 5: Output. Return the top- N words by $p(w \mid T^*)$ as the predicted emerging concepts at time T^* .

For citation recommendation, we rank candidate papers by their contribution to the Hawkes intensity, combining topic relevance $\theta_e[k]$, attention weight $A_{e,\text{ctx}}$, and temporal influence $f_\Delta(T^* - t_e)$.

V. DATASET AND PREPROCESSING

A. Dataset

We use the SciEvo dataset from HuggingFace, consisting of arXiv papers with metadata including titles, abstracts, categories, references, and publication timestamps. We filter papers from two categories: **cs.CL** (Computational Linguistics) and **cs.LG** (Machine Learning), spanning the years 2014-2021.

Table I summarizes the dataset specifications.

TABLE I
DATASET SPECIFICATIONS

Component	Description
Dataset Source	SciEvo (HuggingFace)
Corpus	arXiv cs.CL & cs.LG
Time Range	2014-2021
Topic Modeling	LDA with $K = 30$
Topic Representations	Frozen after LDA training
Training Set	Papers published up to 2020
Test Set	Papers published in 2021

The temporal split is strict: no paper from 2021 is seen during training, preventing any future information leakage.

B. Preprocessing Pipeline

Our preprocessing pipeline consists of the following steps:

- 1) **Data ingestion:** Fetch abstracts and metadata from the SciEvo dataset and Semantic Scholar.
- 2) **Text cleaning:** Lowercasing, removal of URLs, email addresses, numbers, and non-alphabetic characters.
- 3) **Stopword removal:** English stopwords plus domain-specific terms ("paper," "method," "results," "propose,"

etc.) that appear ubiquitously in abstracts but carry no topical information.

- 4) **Lemmatization:** All tokens are lemmatized to reduce vocabulary size (e.g., "learning" and "learned" both map to "learn").
- 5) **Bigram collocation detection:** Identifying meaningful bigrams such as "reinforcement_learning" and "language_model" using statistical co-occurrence measures.
- 6) **LDA training:** Train Gensim LDA with $K = 30$ topics to produce the frozen β (topic-word) and θ (document-topic) matrices.
- 7) **Citation graph extraction:** Construct a sparse paper-to-paper citation adjacency matrix from the reference metadata.
- 8) **Temporal event assembly:** Each paper d becomes an event tuple $(t_d, \theta_d, \mathcal{C}_d)$ containing its publication timestamp, topic vector, and citation list.

VI. EXPERIMENTAL SETUP

We evaluate our model on two complementary tasks:

A. Task 1: Topic Prediction

Given all papers published up to 2020, predict the topic and word distributions for 2021. We compute $\lambda_k(T^*)$ for $T^* = 2021$, convert to topic proportions π_k via softmax, and generate the predicted word distribution $p(w \mid T^*)$. This predicted distribution is compared against the actual empirical distribution of 2021 papers.

Metrics:

- **Pred JSD** (Jensen-Shannon Divergence): Measures distributional distance; lower is better.
- **Pred Cosine:** Cosine similarity between predicted and actual distributions; higher is better.
- **Top-30 Overlap:** Fraction of top-30 predicted words that appear in the top-30 actual words; higher is better.
- **NPMI Coherence:** Measures whether top words within each topic co-occur meaningfully; higher is better.
- **Perplexity:** How well the model predicts held-out words; lower is better.

Baselines: DSNTM [4] and Static LDA.

B. Task 2: Citation Recommendation

Given a paper published in 2021, rank all older papers by their likelihood of being cited. We use the Hawkes intensity contributions to score each candidate.

Metrics:

- **AUC-ROC:** Discrimination between cited and non-cited papers; higher is better.
- **MRR** (Mean Reciprocal Rank): Position of the first correct citation; higher is better.
- **Recall@K:** Fraction of true citations in the top- K recommendations; higher is better.

Baselines: Static LDA (cosine similarity of θ vectors), Content+Recency (cosine similarity plus a recency score), and Uniform Hawkes (recency only, no topic information).

Notably, DSNTM cannot perform citation recommendation because it operates on time-bin-level topic distributions, not individual paper-level scoring.

VII. RESULTS AND ANALYSIS

A. Topic Prediction Results

TABLE II
TOPIC PREDICTION RESULTS

Metric	Hawkes (Ours)	DSNTM	Static LDA
Pred JSD ↓	0.248	0.064	0.226
Pred Cosine ↑	0.923	0.991	0.944
Top-30 Overlap ↑	0.767	0.833	0.800
NPMI Coherence ↑	0.055	0.019	–
Perplexity ↓	2603	1613	–

Analysis. DSNTM outperforms all models on word-level prediction metrics (JSD, Cosine, Top-30 Overlap). This is expected because DSNTM’s time-varying $\beta_k^{(t)}$ adapts vocabulary to each time bin, allowing it to capture vocabulary shifts (e.g., the emergence of “GPT” and “prompt” in recent years). Our frozen β cannot adapt to such shifts.

However, our model produces significantly more coherent topics, with NPMI of 0.055 compared to DSNTM’s 0.019—nearly three times higher. This demonstrates that frozen LDA preserves stable, interpretable topic semantics that joint training in DSNTM disrupts. DSNTM’s optimization for temporal prediction can distort topic boundaries, mixing words from different semantic fields within a single topic.

Static LDA outperforms our model on JSD (0.226 vs. 0.248) and Top-30 Overlap (0.800 vs. 0.767), suggesting that the Hawkes temporal weighting slightly disperses the word distribution—a trade-off for gaining temporal and citation modeling capabilities.

This coherence-versus-prediction trade-off directly motivates our future work on replacing frozen LDA with a Dynamic Topic Model to achieve the best of both worlds.

B. Citation Recommendation Results

TABLE III
CITATION RECOMMENDATION RESULTS

Metric	Ours	Static LDA	Cont.+Rec.	Unif. Hawkes
AUC-ROC ↑	0.845	0.771	0.459	0.390
MRR ↑	0.047	0.044	0.034	0.001
Recall@5 ↑	0.0059	0.0063	0.0048	0.00003
Recall@20 ↑	0.0126	0.0106	0.0115	0.0001

Analysis. Our model achieves the best AUC-ROC (0.845), demonstrating that combining topic relevance, temporal dynamics, and learned attention produces the strongest overall citation ranking.

Uniform Hawkes collapses to near-random (0.390 AUC-ROC, below the 0.5 random baseline), proving that recency alone cannot predict citations—content awareness is essential.

A paper does not cite another paper merely because it was published recently; topical relevance is the primary driver.

Content+Recency also underperforms (0.459), showing that naive content matching without Hawkes-style influence propagation is insufficient. The simple addition of content similarity and recency fails to capture the cascading nature of academic influence.

Static LDA is a strong baseline, achieving 0.771 AUC-ROC and even outperforming our model on Recall@5 (0.0063 vs. 0.0059). At very small K , pure topic similarity is precise: the 5 most topically similar papers are likely to be cited. However, our model overtakes at Recall@20 (0.0126 vs. 0.0106) by surfacing temporally influential papers that topic similarity alone misses—such as foundational optimization papers cited by NLP papers.

The low absolute Recall values across all models are expected given the large candidate pool (thousands of papers) where each paper cites only approximately 20–40 references.

VIII. LIMITATIONS AND FUTURE WORK

While our framework demonstrates strong citation recommendation performance and topic coherence, several limitations and directions for future work remain:

1. Cross-domain interaction weights. Broad CS categories (cs.AI, cs.LG, cs.CL) produce diffuse Hawkes excitation, lowering pinpoint precision. Future work will incorporate explicit cross-domain interaction weights to model dependencies between distinct sub-fields.

2. Neural gating mechanism. The current linear intensity formulation (additive contributions from past events) may be insufficient for capturing sharp, sudden trend bursts. Replacing the linear aggregation with an exponential neural gate could improve forecasting JSD.

3. Dynamic topic representations. Our frozen β cannot capture vocabulary evolution—the word “transformer” carried different meaning in 2014 versus 2021. Shifting from frozen LDA to a Dynamic Topic Model (DTM) would allow word distributions to evolve over time, addressing our primary weakness against DSNTM in word prediction.

4. Paper-specific Weibull parameters. Currently, all papers share global k_w and λ_w . In practice, a comprehensive survey paper has a different citation lifespan than a short workshop paper. Learning paper-specific Weibull parameters could capture these heterogeneous influence patterns.

IX. CONCLUSION

We presented a two-stage framework for temporal topic modeling and citation recommendation that combines frozen LDA topic representations with a Hawkes process augmented by citation-supervised attention and a Weibull temporal kernel. Our approach explicitly separates topic discovery from temporal dynamics, yielding interpretable and coherent topics while enabling continuous-time influence modeling.

On citation recommendation, our model achieves the best AUC-ROC (0.845), significantly outperforming all baselines and demonstrating the value of combining topic relevance,

learned attention, and temporal dynamics. On topic prediction, while DSNTM’s time-varying vocabulary gives it an edge in word-level metrics, our model produces nearly three times more coherent topics (NPMI 0.055 vs. 0.019).

The trade-off between frozen and dynamic topic representations highlights a fundamental tension in temporal topic modeling: stability and interpretability versus vocabulary adaptability. Our future work on integrating dynamic topic models with the Hawkes framework aims to resolve this tension, achieving the best of both approaches.

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